

SIDDHARTHAN L

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PROFILE

Data Science postgraduate specializing in Data Analysis, with hands-on experience in Python, SQL, Power BI, Excel, and MySQL for data analysis, statistical modeling, and business intelligence. Skilled in data cleaning, data wrangling, exploratory data analysis (EDA), predictive modeling, and machine learning using Pandas, NumPy, and Scikit-learn, with additional exposure to natural language processing (NLP) and large language models (LLMs) through Retrieval-Augmented Generation (RAG) systems. Experienced in building predictive models, interactive dashboards, and data pipelines that convert raw data into actionable, data-driven business insights. Seeking a Data Analyst or Data Scientist role to apply analytical, statistical, and machine learning skills to solve real-world business problems.

SKILLS

Programming Languages: Python, SQL

Data Analysis & Wrangling: Pandas, NumPy, Data Cleaning, Data Wrangling, Data Preprocessing, Exploratory Data Analysis (EDA)

Statistics & Machine Learning: Statistical Analysis, Predictive Modeling, Regression, Classification, Clustering, Feature Engineering, Model Evaluation (Accuracy, F1-Score, AUC-ROC, Recall), Time Series Forecasting (ARIMA), Ensemble Methods, SMOTE / Imbalanced Data Handling

Data Visualization & BI: Power BI, Microsoft Excel (Pivot Tables, VLOOKUP), Matplotlib, Seaborn, Dashboard Development, KPI Reporting, Data Storytelling

Databases & Querying: MySQL, SQL (Joins, Subqueries, Window Functions, Aggregate Functions)

NLP & Generative AI: LangChain, FAISS, Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Vector Embeddings

Tools & Platforms: Jupyter Notebook, Streamlit, Git/GitHub

PROJECTS

Online Payment Fraud Detection Using Machine Learning

Technologies: Python, Pandas, Scikit-learn, Matplotlib

- Engineered and benchmarked three classification models (Logistic Regression, Decision Tree, Random Forest) on a 50,000+ record transaction dataset to detect fraudulent payments, applying feature engineering to strengthen predictive performance.
- Resolved severe class imbalance by applying SMOTE resampling, raising fraud detection recall from 62% to 89%.
- Delivered a Random Forest model achieving 96.2% accuracy, a 0.90 F1-score, and a 0.97 AUC-ROC through rigorous model evaluation, providing a reliable fraud-flagging solution.
- Conducted exploratory data analysis (EDA) and statistical analysis to identify key fraud indicators, sharpening feature selection and model interpretability.

AI-Powered Retrieval-Augmented Generation (RAG) Chatbot for PDF Document Analysis

Technologies: Python, Streamlit, LangChain, FAISS

- Architected an end-to-end Retrieval-Augmented Generation (RAG) data pipeline — PDF parsing, text chunking, vector embeddings, and FAISS vector database indexing — to enable semantic search across documents exceeding 200 pages.
- Built a Streamlit-based chatbot interface using LangChain and large language models (LLMs), sustaining an average query response time under 3 seconds across 50+ documents.
- Reduced manual document search effort by approximately 70% by replacing manual browsing with natural language querying (NLP).

Gold Price Forecasting Using Time Series Analysis and Machine Learning

Technologies: Python, Pandas, ARIMA, Statsmodels, Matplotlib

- Developed an ARIMA-based time-series forecasting model using 10+ years of historical gold price data to project future price movements.
- Performed data preprocessing, trend analysis, and seasonal decomposition to strengthen forecasting accuracy.
- Achieved a 4.5% Mean Absolute Percentage Error (MAPE), outperforming a baseline forecasting approach.
- Visualized historical trends and 30-day price forecasts to support data-driven decision-making for stakeholders.

EDUCATION

Master of Science in Data Science

Vellore Institute of Technology, Vellore | CGPA: 8.09 2024 – 2026

Bachelor of Science in Computer Science

C. Abdul Hakeem College, Ranipet | CGPA: 8.0 2021 – 2024

CERTIFICATIONS

- Business Intelligence and Analytics — NPTEL [E-CERTIFICATE](#)